

# BobaLink “ User Guide

**Revision:** 1 **Last modified:** 2026-06-10T21:00:00Z **Scope:** End-user guide for BobaLink, the Boba Project browser extension (`extension/`). **Authority:** `docs/browser_extension/Status.md` (Rev 4) + `docs/browser_extension/IMPLEMENTATION_PLAN.md`.

Accuracy note (Â§11.4.6): every statement below reflects the real extension source/config as of HEAD 2011810. Capabilities still in progress are marked **(planned)** or **(in progress)** so nothing reads as shipped that is not.

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## What BobaLink does

BobaLink is a cross-browser **Manifest V3** WebExtension (built with WXT + TypeScript). It:

- **detects** magnet links and `.torrent` file URLs on the pages you visit;
- **parses** them (magnet bti h infohash / `.torrent` “ SHA-1 infohash) and **deduplicates** by infohash so the same torrent is never sent twice;
- **forwards** them to the **Boba merge service** on `http://localhost:7187`, which adds them to qBittorrent server-side.

The extension never talks to qBittorrent directly “ it only knows about the Boba merge service on port 7187. (`extension/wxt.config.ts host_permissions = http://localhost:7187/*`; `extension/src/shared/constants.ts DEFAULT_URLS`.)

The end-to-end detect “ send “ torrent-appears-in-qBittorrent path against a **live** backend is **(in progress)** “ Phase 4 in the plan. The detection, parsing, dedup, and UI layers are implemented and tested.

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## Supported torrent sites

Detection works on **any page** via the generic `magnet:` and `.torrent` link patterns. In addition, BobaLink ships a curated selector table (`SITE_SELECTORS` in `extension/src/shared/constants.ts`) tuned for these hosts. The content script's matches are derived from this table “ BobaLink does **not** request `<all_urls>`.

The curated table covers **24 entries** (one generic pattern set plus 23 site-specific hosts):

| # | Host               | Notes                                                            |
|---|--------------------|------------------------------------------------------------------|
| “ | generic            | magnet / .torrent / download.php?id= patterns applied everywhere |
| 1 | 1337x.to           |                                                                  |
| 2 | thepiratebay.org   |                                                                  |
| 3 | thepiratebay10.org |                                                                  |
| 4 | rarbg.to           |                                                                  |
| 5 | rarbgtorrents.org  |                                                                  |
| 6 | yts.mx             |                                                                  |
| 7 | yts.lt             |                                                                  |

| #  | Host                 | Notes                |
|----|----------------------|----------------------|
| 8  | eztv.re              |                      |
| 9  | limetorrents.lol     |                      |
| 10 | torrentgalaxy.to     |                      |
| 11 | nyaa.si              | anime                |
| 12 | animetosho.org       |                      |
| 13 | torrentz2.eu         |                      |
| 14 | fitgirl-repacks.site |                      |
| 15 | rutracker.org        | Boba private tracker |
| 16 | kinozal.tv           | Boba private tracker |
| 17 | nnmclub.to           | Boba private tracker |
| 18 | rutor.info           | Boba public tracker  |
| 19 | katcr.co             | Kickass Torrents     |
| 20 | demonoid.is          |                      |
| 21 | iptorrents.com       | Boba private tracker |
| 22 | torrentleech.org     | private              |
| 23 | beyond-hd.me         | private              |
| 24 | passthepopcorn.me    | private              |

(That is 23 site-specific hosts plus the generic set â€” the project refers to this as the **24 supported sites** table.)

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## How detection works

- The content script scans the page for magnet links and .torrent URLs using the site selectors above plus the generic regex patterns (MAGNET\_REGEX, TORRENT\_FILE\_REGEX in constants.ts).
- It watches the page for changes with a MutationObserver (500 ms debounce â€” DEBOUNCE\_DELAYS.MUTATION) so links added by single-page-app navigation are picked up too.
- Each detected item is parsed to its infohash and **deduplicated by lowercase infohash**, so the same torrent appearing in several places on a page is sent once.

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## The popup

Click the BobaLink toolbar icon to open the popup. It shows the torrents detected on the current page and lets you send them to Boba:

- **Send** â€” send a single selected detected item to the Boba merge service.
- **Send All** â€” send every detected item on the current page (deduplicated).

The popup also surfaces a connection/status indicator (badge colours are defined in BADGE\_COLORS: healthy / degraded / error / scanning / detected).

The live-backend round-trip these buttons perform is **(in progress)** â€” Phase 4. The popup, detected-list rendering, and Send / Send-All wiring are implemented and tested.

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## Context-menu actions

Right-click a page (or a magnet link) to use BobaLink's context-menu items (defined in `extension/src/background/index.ts`):

- **Send magnet to Boba** (`bobalink-send`) â€” send the right-clicked magnet link.
- **Send all on page** (`bobalink-send-all`) â€” send every detected torrent on the current page.
- **Send tab group** (`bobalink-send-group`) â€” scan **every tab in the current tab group**, deduplicate across the whole group, and send them as one batch.

The "Send tab group" batch path is wired (`MENU_SEND_GROUP` â†’ deduped group batch â†’ a single `addMagnets` request). It is a Chrome-family feature (uses `chrome.tabGroups`); on browsers without tab groups it degrades gracefully.

You can hide the context-menu items from **Options** â†’ **UI** â†’ **Show context-menu items**.

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## Keyboard shortcuts

Three shortcuts are declared in `extension/wxt.config.ts` (commands):

| Action                                         | Windows / Linux macOS        |
|------------------------------------------------|------------------------------|
| Send to Boba ( <code>send-to-boba</code> )     | Ctrl+Shift+B Command+Shift+B |
| Scan page ( <code>scan-page</code> )           | Ctrl+Shift+S Command+Shift+S |
| Open dashboard ( <code>open-dashboard</code> ) | Ctrl+Shift+D Command+Shift+D |

These are *suggested* keys; your browser may let you remap them on its extension shortcuts page (e.g. `chrome://extensions/shortcuts`). You can disable shortcut handling from **Options** â†’ **UI** â†’ **Enable keyboard shortcuts**.

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## The 7 options tabs

Open the options page (right-click the icon â†’ Options, or your browser's extension-details "Extension options"). It has **seven tabs** (`extension/src/entrypoints/options/index.html` + `src/options/options.ts`, `TAB_IDS`):

1. **Server** â€” server name, **Server URL** (default `http://localhost:7187`), authentication method (None / Username+Password cookie / API Key / Basic), the optional **Boba API token** + **Session passphrase** (see below), connection timeout (seconds), health-check interval (minutes), and "Verify HTTPS certificates (HTTPS only)".
2. **Download Prefs** â€” default category, default save path, content layout (Original / Create subfolder / No subfolder), pause-after-add, skip hash check, automatic torrent management.
3. **Queue** â€” enable the offline queue for failed sends, max queue size (default 50), max history items (default 100).

4. **Notifications** â€” show notifications, play a sound on completion, auto-send detected torrents (no confirmation).
  5. **Detection** â€” automatically scan pages, auto-scan delay (ms), highlight detected torrents, highlight style (Badge / Border / Glow).
  6. **UI** â€” show context-menu items, enable keyboard shortcuts.
  7. **Security** â€” explains the delegate-by-default model and offers an "Enable debug logging" toggle.
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## The optional Boba API token + session passphrase

BobaLink follows a **delegate-by-default** security model: in the normal localhost deployment it stores **no decryptable secret at all** â€” it just posts magnets/URLs to the local merge service, which already owns the qBittorrent and tracker credentials.

The **only** optional secret is a **Boba API token**, which is needed *only if* your Boba backend sets the BOBA\_API\_TOKEN environment variable (the backend token gate is open by default and enforced only when that variable is set).

When you supply a token on the **Server** tab, BobaLink encrypts it **only under a session passphrase you type** (AES-GCM-256 via PBKDF2 â€” ENCRYPTION in `constants.ts`):

- The passphrase is held in `chrome.storage.session` only and is **never persisted** to disk.
- There is **no fixed/embedded encryption key and no empty-passphrase path** â€” if you enter a token but no passphrase, the token is **not stored** and a notice tells you to add a passphrase (`src/options/options.ts`).
- The plaintext token and the passphrase are never logged.

When the token is present and unlocked, the Phase-4 client sends it as either an `Authorization: Bearer <token>` or `X-Boba-Token: <token>` header (BOBA\_TOKEN\_HEADERS). Full decrypt-and-send wiring is landed and tested; the end-to-end against a live token-gated backend is **(in progress)**.

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## Least-privilege permissions â€” what BobaLink asks for and why

BobaLink requests only what it needs (`extension/wxt.config.ts`):

| Permission    | Why it is requested                                                   |
|---------------|-----------------------------------------------------------------------|
| storage       | persist your settings, detected items, the offline queue, and history |
| alarms        | keep the service worker alive and run periodic health checks          |
| notifications | tell you when a send succeeds/fails                                   |
| activeTab     | act on the page you explicitly invoke BobaLink on                     |
| contextMenus  | provide the right-click "Send â€”" actions                            |
| tabGroups     | read which group the clicked tab belongs to, for "Send tab group"     |

## Permission

`host_permissions: http://  
localhost:7187/*`

## Why it is requested

talk to the local Boba merge service â€”  
nothing else

## What BobaLink deliberately does not request

- `tabs` â€” the tab-group batcher reads only `tab.id`, which per the Chrome docs needs no `tabs` permission. BobaLink never reads tab URLs/titles/favicons.
- `scripting` â€” not requested.
- `cookies` â€” not requested; BobaLink never touches your cookies.
- `<all_urls>` â€” the content script's match patterns are derived from the curated `SITE_SELECTORS` table, not a blanket all-sites grant.

Its content security policy is tight: `script-src 'self'` (no `unsafe-inline` / `unsafe-eval`), and `connect-src` scoped to `http://localhost:7187` (`content_security_policy` in `wxt.config.ts`).

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## Internationalization & accessibility

- BobaLink ships UI message catalogs for **English** (`en`), **Russian** (`ru`), **German** (`de`), and **French** (`fr`) (`src/public/_locales/{en,ru,de,fr}`, packaged into the build). Chrome/Firefox auto-select by the browser UI language, falling back to English. More locales are **(planned)** (the plan targets 8).
  - The popup and options UI carry ARIA roles, accessible names, and `tablist/tabpanel` wiring, validated by an accessibility test suite. Deeper WCAG work (contrast/focus/full keyboard coverage) and theme-switch evidence are **(in progress)**.
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## Current status at a glance

Phases 1â€”3 (foundation, detection/parsing, the content/background/popup/options shell) plus the WXT build wiring are **PASS**. Backend integration (Phase 4), tab-group batch polish (Phase 5), `i18n/ally/themes` (Phase 6), the full security/credentials suite (Phase 7), full multi-type test coverage (Phase 8), and distribution (Phase 9) are **in progress**. See `docs/browser_extension/Status.md` for the authoritative per-phase state.